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From curve to equation: how the Lotus Temple was made buildable

The free guide to the reel on the Lotus Temple in Delhi: why a flower-shaped building needs equations, how engineers turn free forms into simple geometry, and how to try it on your own sketch.



How it works

A free curve is easy to sketch and hard to build. A builder needs numbers: where a surface starts, how far it curves, where it ends. The trick used on complex buildings is called rationalisation: describe the form as a set of simple, well-known surfaces whose shape can be written as an equation.

Once each part is an equation, it can be checked structurally, and it can be turned into hundreds or thousands of exact coordinates for the people building the formwork. The architect keeps the idea, the engineer gets numbers.

A guide to the Lotus Temple describes the surfaces as simple solids such as spheres, cylinders, cones and rings. This is one source, so treat the exact list as indicative.

The Lotus Temple in facts

The Bahá'í House of Worship in New Delhi was designed by Fariborz Sahba. It was dedicated on 24 December 1986 and opened to the public on 1 January 1987. It is 40.8 m high, about 70 m across, and seats 1,300.

It has 27 free-standing petals of cast-in-place reinforced concrete clad in white marble from Penteli, Greece, in three rings of nine. Nine appears everywhere in the design, linked to the number's meaning in the Bahá'í Faith. The London engineers Flint & Neill spent about 18 months turning the concept into geometry that could be calculated and built. The contractors were Larsen & Toubro's ECC group.

Built examples

Lotus Temple, New Delhi (1986)	Fariborz Sahba with Flint & Neill: 27 concrete shells, geometry resolved for calculation and setting out.
Sydney Opera House (1973)	Utzon and Arup: the shells were rationalised as parts of a sphere, so one mould could make many pieces.
Guggenheim Bilbao (1997)	Gehry's team digitised models and cut the skin into panels (see the Bilbao guide).
Eden Project Core, Cornwall (2005)	Grimshaw: roof geometry organised by a spiral pattern from plant growth.

Try it yourself

1. Draw a free curved form on paper: a leaf, a petal, a wave.
2. Lay tracing paper over it and cover the curve with circle arcs: find the circle that best fits each part.
3. Write down the radius and the angle of each arc.
4. Rebuild the shape in a free tool: draw the arcs in Geogebra or Desmos using your numbers.
5. Compare your rebuilt shape with the original and note where it differs.
6. Try fewer arcs, then more: how many pieces do you need before the difference disappears to the eye?
7. Cut the arcs from card and join them: could a factory repeat any of them?

The rule to remember

Next time a form looks impossible, ask which simple shapes it hides.

Starting from the curve	Start from a family of simple surfaces and shape the design inside it.
Too many different pieces	Fewer repeated parts make formwork cheaper.
Skipping the engineer	Bring the engineer in while the geometry is still open.

Go further

Lotus Temple, Wikipedia
https://en.wikipedia.org/wiki/Lotus_Temple

Lotus Temple / Fariborz Sahba, Classics on Architecture Lab
https://www.architecturelab.net/lotus-temple-fariborz-sahba/

Architectural blossoming of the lotus, Bahá'í Library (Naharoy)
https://bahai-library.com/pdf/n/naharoy_architectural_blossoming_lotus.pdf

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